

Acoustics for Engineers

Troy Lectures

Jens Blauert and Ning Xiang

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Second Edition



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Preface

This book provides the material for an introductory course in engineering acoustics for students with basic knowledge in mathematics. It is based on extensive teaching experience at the university level.

Under the guidance of an academic teacher it is sufficient as the sole textbook for the subject. Each chapter deals with a well defined topic and represents the material for a two-hour lecture. The chapters alternate between more theoretical and more application-oriented concepts.

For the purpose of self-study, the reader is advised to use this text in parallel with further introductory material. Some suggestions to this end are given in Appendix 15.3.

The authors thank *Dorea Ruggles* for providing substantial stylistic refinements. Further thanks go to various colleagues and graduate students who most willingly helped with corrections and proof reading and, last but not least, to the reviewers of the 1st edition, particularly to Profs. *Gerhard Sessler* and *Dominique J. Chéenne*. Nevertheless, the authors assume full responsibility for all contents.

For the 2nd edition, typos have been corrected and a number of figures, notations and equations have been edited to increase the clarity of presentation. Further, a collection of problems has been included. Solutions to the problems will be provided on a peer-to-peer basis via the internet – see Appendix 15.4 for the link.

Bochum and Troy,
June 2009

Jens Blauert
Ning Xiang

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